

1. What is the area of a circle with a radius of 7 cm? (Use $\pi = 22/7$)

a) 154 cm²

b) 49 cm²

c) 77 cm²

d) 100 cm²

2. A person invests \$2000 at 5% annual interest compounded annually. What will be the amount after 2 years?

a) \$2100

b) \$2205

c) \$2300

d) \$2500

3. A bag contains 3 red balls, 5 blue balls, and 2 green balls. What is the probability of picking a blue ball from the bag?

a) 5/10

b) 5/7

c) 3/10

d) 2/7

4. A car travels 180 km in 3 hours. What is its average speed?

a) 50 km/h

b) 55 km/h

c) 60 km/h

d) 65 km/h

5. A sum of money doubles itself in 5 years at simple interest. What is the rate of interest?

a) 10%

b) 12%

c) 15%

d) 20%

6. If 6 men can complete a task in 8 days, how many days will 9 men take to complete the same task?

a) 6 days

b) 5 days

c) 7 days

d) 4 days

7. A man is paid \$5 per hour for the first 40 hours and \$7 per hour for each hour over 40 hours. How much will he earn if he works 50 hours?

a) \$350

b) \$400

c) \$375

d) \$270

8. What is the value of $(x + 3)^2 - (x - 2)^2$?

a) $5x + 6$

b) $x^2 + 6x - 6$

c) $10x + 5$

d) $6x + 6$

9. If a man can finish a task in 10 days, how many men will be required to finish the task in 5 days?

a) 12

b) 15

c) 18

d) 20

10. What is the greatest common divisor (GCD) of 28 and 42?

a) 6

b) 7

c) 10

d) 14

11. If a fraction is $\frac{4}{5}$, and the numerator is increased by 2, the fraction becomes $\frac{3}{2}$. What is the original denominator?

a) 4

b) 6

c) 7

d) 8

12. What is the value of $8!$ (8 factorial)?

a) 40320

b) 362880

c) 5040

d) 840

13. The price of a shirt is increased by 25%. If the original price was \$500, what is the new price?

a) \$600

b) \$625

c) \$700

d) \$550

14. A number is increased by 40%. If the new number is 140, what was the original number?

a) 100

b) 90

c) 80

d) 110

15. If the perimeter of a rectangle is 36 meters and its length is 10 meters, what is the width?

a) 5 meters

b) 6 meters

c) 7 meters

d) 8 meters

16. If $x = 3$, what is the value of $5x^2 - 4x + 7$?

a) 30

b) 35

c) 40

d) 45

17. The sum of the ages of A and B is 40 years. If A is 6 years older than B, what is B's age?

a) 16

b) 17

c) 18

d) 19

18. If a number is increased by 25%, it becomes 500. What is the original number?

a) 375

b) 400

c) 425

d) 450

19. If a number is divided by 3, the remainder is 2. What is the number?

a) 5

b) 8

c) 11

d) 14

20. A sum of money is invested at 10% per annum simple interest for 4 years. If the total interest is \$200, what is the principal amount?

a) \$500

b) \$400

c) \$1000

d) \$600

21. A train 200 meters long is running at a speed of 60 km/h. How much time will it take to cross a man standing on the platform?"

a) 10 seconds

b) 12 seconds

c) 15 seconds

d) 20 seconds

22. A shopkeeper gives a discount of 20% on the marked price. If the cost price is ₹400 and he still gains 25%, what is the marked price?

a) ₹1,800

b) ₹1,600

c) ₹1,500

d) ₹2,000

23. What does the command `chmod 755 file` do in Linux?

a) Gives the file only read and write permissions

b) Sets the file as executable for the owner

c) Assigns read, write, and execute for the owner; read and execute for others

d) Deletes the file

24. What happens if the condition in a while loop is always true?

a) The loop executes once

b) The loop will result in an error

c) The loop becomes an infinite loop

d) The loop does not execute

25. In SQL, which clause is used to filter rows based on aggregate functions?

a) WHERE

b) GROUP BY

c) HAVING

d) ORDER BY